

Multimodal Music Recommendation System using LLMs

Srikar Prabhas Kandagatla¹, Sreehitha Narayana¹, Swetha Mohan¹, Chandana Magapu¹, Shamanth Kuthpadi Seethakanth¹, Hongjie Chen², Ryan Rossi³, Franck Dernoncourt³

¹University of Massachusetts Amherst, ²Dolby Labs, ³Adobe

Abstract

Music recommendation systems typically treat songs as opaque tokens, relying on collaborative interaction histories which overlooks semantic or acoustic content. While recent LLM-based recommenders improve sequential reasoning by leveraging user behavior, they remain largely ID-dependent with limited grounding in actual song content. Prior work has explored LLM-augmented, multimodal, and text-enhanced approaches to sequential recommendation, but none jointly model semantic, acoustic, and engagement signals for music. In this work, we propose a multimodal framework for session-based music recommendation that enriches the LastFM-1K dataset with three complementary signals: (1) LLM-generated semantic metadata using the MGPHot annotation schema, (2) audio embeddings from pretrained music representation models, and (3) listening completion ratios. We integrate these features into E4SRec and evaluate multiple fusion strategies under zero-shot and fine-tuned settings. Our experiments show that semantic and acoustic enrichment improves over ID-only baselines by 5–20%, with lyric-aware semantic representations producing the most consistent gains up to 24%. Naive multimodal fusion does not always yield additive improvements, highlighting challenges in cross-modal integration. We release a large-scale multimodal benchmark for music recommendation.

Keywords

Music Recommendation, Multimodal Learning, Large Language Models, Sequential Recommendation, Audio Embeddings, Metadata Enrichment

1 Introduction

Sequential music recommendation aims to predict the next song a user is likely to listen to based on their listening history [31, 35, 37, 43]. Modern recommender systems commonly formulate this task using sequential recommendation architectures, where songs are represented as discrete item identifiers and user behavior is modeled as sequences of item interactions [7, 32, 38]. While these ID-based approaches perform well in dense interaction settings, they treat each track as an independent token and fail to capture meaningful relationships between songs [16, 19, 40]. As a result, they struggle under sparsity and cold-start conditions and cannot effectively reason about acoustic similarity, lyrical themes, or semantic context.

Music preference is inherently multimodal, influenced by acoustic properties such as tempo and timbre, lyrical content, and higher-level semantic attributes including mood, instrumentation, and genre. Recent work has explored the use of large language models (LLMs) for recommendation due to their ability to model sequential behavior and reason over textual inputs. However, many existing LLM-based recommenders still rely heavily on ID-based embeddings or weak textual proxies, limiting their grounding in the actual

content of songs. Consequently, recommendations may lack semantic coherence or fail to reflect meaningful similarity between tracks.

Recent studies suggest that incorporating multimodal information can produce richer item representations and improve recommendation quality. For example, TalkPlay [4] demonstrates that grounding LLM-based music recommenders with audio, lyrics, and semantic metadata improves both recommendation relevance and conversational quality. Despite these advances, multimodal grounding has not been systematically explored within sequential music recommendation frameworks.

In this work, we address this gap by constructing a multimodal music recommendation framework on top of the LastFM-1K dataset [6]. We enrich song representations with three complementary modalities: (1) LLM-generated semantic metadata describing musical characteristics such as harmony, rhythm, sonority, instrumentation, and lyrical themes, (2) audio-derived representations including handcrafted acoustic descriptors and pretrained audio embeddings, and (3) behavioral engagement signals derived from listening completion ratios. These enriched representations are integrated into sequential recommendation models including SASRec [11] and E4SRec [12] to evaluate how different modalities contribute to recommendation performance.

Our contributions are:

- We identify limitations of ID-based sequential music recommendation approaches that lack semantic and acoustic grounding.
- We construct a multimodal benchmark by enriching the LastFM-1K dataset with LLM-generated semantic metadata, audio representations, and behavioral engagement signals.
- We propose a framework for integrating multimodal representations into sequential recommendation models through multiple fusion strategies.
- We perform controlled experiments across different recommendation backbones and modality combinations, showing that multimodal enrichment improves recommendation quality while highlighting challenges in modality fusion.

2 Related Work

2.1 Sequential Recommendation

Sequential recommendation (SR) shifts personalization away from static profiles to model the dynamic evolution of user preferences using the chronological order of historical behaviors. Recent advances in Transformers, Graph Neural Networks, and LLM-based recommenders have significantly improved the ability of SR models to capture long-range dependencies and multimodal user behavior.

Li et al. [12] proposed *E4SRec*, which integrates LLMs with ID-based sequential recommenders by directly injecting pretrained item ID embeddings into the input layer and replacing autoregressive generation with a single-pass prediction layer. Using LoRA

for efficient adaptation, E4SRec achieves state-of-the-art accuracy while mitigating data sparsity without relying on rich semantic descriptions. Wang et al. [34] proposed *IRLLRec*, a model-agnostic framework that bridges the semantic gap between unstructured text and structured interactions via a dual-tower architecture with pairwise and translation alignment strategies. An Interaction-Text Matching module further distills fine-grained latent preferences across modalities, significantly boosting both performance and interpretability over collaborative filtering baselines. Tran et al. [30] proposed *PISA*, a Transformer-based sequential recommender tailored for music listening that integrates ACT-R cognitive memory mechanisms to capture repetitive listening behavior. Evaluations on Last.fm and Deezer demonstrate that PISA accurately predicts engagement with both novel and previously heard tracks. Du et al. [5] proposed *IntervallLM*, which incorporates irregular time intervals into LLMs via an Interval-Infused Attention mechanism, outperforming baselines by an average of 4.4% across overall, warm, and cold-start scenarios.

2.2 Session-based Recommendation

Session-based recommendation predicts the next item using only interactions within a short user session, without relying on long-term user profiles. This setting is particularly important in music streaming, where user intent shifts rapidly and contextual listening behavior strongly influences track selection.

Yu et al. [39] proposed *TAGNN*, a graph-based session recommendation model that represents each session as a directed graph and introduces a target-aware attention mechanism that computes candidate-specific session representations. By dynamically attending to different parts of the session depending on the target item, TAGNN achieves more expressive next-item prediction, particularly relevant for music sessions containing multiple micro-intents. Hansen et al. [8] study large-scale music recommendation in real-world streaming systems, learning dynamic user representations that evolve based on recent listening activity. Their results demonstrate that accounting for temporal dynamics significantly improves next-track prediction at the production scale. While these works effectively capture behavioral signals from interaction sequences, neither incorporates richer semantic representations beyond collaborative signals. Our approach extends session-based modeling by integrating LLM-generated metadata and audio embeddings, enabling deeper understanding of song characteristics within evolving listening sessions.

2.3 Multimodal Recommendation

Multimodal recommendation extends traditional recommender systems by incorporating heterogeneous content signals such as text, audio, images, and structured metadata alongside interaction data.

Qin [26] proposed *ATFLRec*, an audio-text fusion framework built on an instruction-tuned LLM that jointly models cross-modal interactions within the LLM space using LoRA for efficient adaptation, showing consistent improvements over unimodal and graph-based baselines. Unlike ATFLRec, which assumes direct access to raw audio, our approach uses externally derived audio embeddings, enabling scalable multimodal modeling without raw waveform access. Lyu et al. [20] proposed *X-Reflect*, a cross-reflection prompting

framework that explicitly prompts MLLMs to reconcile supportive and conflicting signals between text and images, producing more nuanced item representations and outperforming existing prompting baselines. Pomo et al. [25] show that naive fusion strategies often fail to achieve meaningful cross-modal alignment, and that Large Vision-Language Models with structured prompting produce semantically coherent embeddings that outperform simple feature concatenation, directly motivating our use of LLM-generated metadata alongside audio embeddings. Overall, effective cross-modal alignment is critical for recommendation robustness, yet most existing approaches focus on static item enrichment and assume full modality access. Our work addresses both limitations by integrating externally derived audio embeddings with LLM-generated metadata within a session-aware framework.

2.4 Representation for Music Recommendation

Accurate user and item representations are the cornerstone of modern music recommendation systems. By mapping sparse interaction signals into dense vectors, a recommender can measure similarity, compute relevance scores, or feed downstream generative models. We summarize the main families of embeddings explored for music recommendation and their impact on recommendation quality.

The most traditional approach learns a low-dimensional vector per user and per track directly from the interaction matrix. Methods such as BPR [28], LightGCN [10], and NGCF [33] capture pure collaborative signals and yield strong ranking results on dense datasets, but performance degrades sharply under data sparsity or cold-start conditions due to tight coupling with observed interactions. To alleviate sparsity, many works augment or replace item ID vectors with dense acoustic representations extracted from the raw waveform or spectrogram, using back-ends such as *MusiCNN* [2], *MERT* [15], *Music2Vec*, *MusicFM*, and *EncodecMAE* [13, 14, 41]. These item-side embeddings are either frozen or fine-tuned jointly with the recommendation loss, and empirically improve Recall and NDCG by 5–15% on sparse playlists [29], especially when combined with collaborative ID vectors. When metadata such as titles, lyrics, tags, or cover art is available, BERT/Roberta-style models and ResNet/ViT encoders can supply rich item-side vectors, while systems like CLAP [9] align audio and text in a shared space for cross-modal retrieval. Studies report consistent gains of 3–10% in NDCG over pure ID baselines, particularly for cold-start items [18]. Item attributes and relational facts can be modeled with KG techniques such as TransE, RGCN, or CompGCN and merged with collaborative vectors. The KDR framework [22] aligns explicit KG embeddings with implicit collaborative factors via mutual-information maximisation, yielding interpretable dimensions (e.g., “genre”, “tempo”) and up to 8% improvement in HR@1 on music datasets. A newer line of work factorises embeddings into sub-spaces corresponding to orthogonal musical factors such as mood, instrumentation, and tempo. Matryoshka Representation Learning [42] creates a hierarchy of increasingly fine-grained vectors, while KDR aligns disentangled factors with KG semantics; both approaches mitigate over-generalisation and improve diversity-aware and long-tail recommendation metrics.

In summary, representation learning for music recommendation has progressed from pure ID-based collaborative vectors to

Table 1: LastFM-1K dataset statistics after preprocessing.

Metric	Value
Number of users	814
Total scrobbles	4,213,960
Unique tracks	295,957
Unique artists	43,406
Number of sessions	421,396
Mean session duration (mins)	44.39
Median session duration (mins)	41.33
Sparsity	0.0175
Date range	2005-02-14 to 2009-06-19

rich, multimodal, and semantically structured embeddings, with a direct and measurable effect on recommendation quality, especially under data sparsity or cold-start scenarios. These findings motivate our multimodal recommendation framework, which integrates collaborative, semantic, and acoustic representations within a session-based recommendation setting.

3 Benchmark Construction

We construct a benchmark dataset for multimodal music recommendation by enriching the publicly available LastFM-1K user interaction dataset with a curated catalog of song-level features drawn from multiple data sources.

3.1 Interaction Dataset and Sessions

We build our benchmark on top of the **LastFM-1K** dataset [6], a publicly available collection of timestamped listening histories from approximately 1,000 users of the Last.fm music streaming platform. Each interaction record contains a user identifier, timestamp, artist name, track name, and associated MusicBrainz identifiers.

To construct a session-based recommendation dataset, we filter users with fewer than 1,000 listening events and tracks with fewer than 7 total plays. Following prior work on session-based music recommendation [30], we segment user histories into sessions by splitting interactions separated by more than 20 minutes of inactivity. Sessions containing fewer than 10 interactions are discarded. The resulting dataset contains 814 users, 4.2 million listening events, and 295,957 unique tracks spanning February 2005 to June 2009. Table 1 summarizes the statistics of the processed interaction dataset.

To support multimodal recommendation experiments, we further construct a Top-50k song catalog by selecting the 50,029 most frequently played tracks from the filtered interaction data. Each track is subsequently enriched with semantic, acoustic features described in the following subsections.

3.2 Musicological Features

To capture structured musicological properties, we annotate songs using the MGPHot schema [23], which defines 58 musicological attributes spanning lyrics, vocals, harmony, rhythm, instrumentation, sonority, and composition. These attributes provide interpretable descriptions of musical characteristics that are not directly observable from collaborative interaction histories alone.

Table 2: MGPHot attribute categories and representative features.

Category	Representative attributes
Lyrics	Angry, Sad, Happy, Humorous, Love (mood dimensions)
Vocals	Register, Timbre (Thin-Full), Breathiness, Smoothness
Harmony	Minor/Major Key Tonality, Harmonic Sophistication
Rhythm	Tempo, Swing Feel, Syncopation, Danceability
Instrumentation	Drum Set, Electric Guitar, Acoustic Guitar, Piano
Sonority	Live Recording, Acoustic, Electric
Composition	Focus on Lead Vocal, Focus on Melody, Focus on Lyrics

For each track, we query an LLM (Azure OpenAI GPT-4o, via the Batch API) with the artist name and track title and ask it to rate every MGPHot attribute on a Likert-type scale from 0 (absent / lowest) to 5 (dominant / highest). After validating on a held-out subset of the original MGPHot annotations, we selected a *few-shot* prompting variant that prepends two multi-turn anchor examples whose ground-truth ratings are taken directly from the published MGPHot annotations.

The two anchor songs were chosen deliberately to represent stylistic extremes that span the full 0–5 rating range across all 58 attributes. Simon & Garfunkel’s *Mrs. Robinson* represents an acoustic, folk-influenced track with prominent clean vocals, minimal electric instrumentation, and high lyrical coherence, while Metallica’s *Master of Puppets* represents the opposite extreme: heavy electric guitars, aggressive vocals, complex rhythmic structures, and high sonority intensity. Together, these two anchors provide the model with calibration examples that cover both ends of the rating scale for the most discriminative attributes, including vocal register, timbre, breathiness, syncopation, and electric instrumentation. These anchors calibrate the model’s use of the 0–5 range across stylistic extremes. The full prompt template is shown in Figure 1. Raw integer ratings are subsequently rescaled to $[0, 1]$ by dividing by 5, yielding continuous-valued feature vectors that describe properties such as vocal timbre, harmonic sophistication, rhythmic complexity, instrumentation profile, emotional tone, and production style.

Unlike traditional categorical metadata, MGPHot features provide dense semantic representations that can model gradual variation between songs. For example, rhythmic attributes such as swing feel or syncopation are represented on continuous scales rather than binary labels, enabling finer-grained similarity modeling within recommendation systems.

We group the extracted attributes into seven high-level categories: Lyrics, Vocals, Harmony, Rhythm, Instrumentation, Sonority, and Composition. Table 2 summarizes the feature groups and representative attributes used in our benchmark.

To ensure consistency and scalability across the large song catalog, all annotations are generated through the standardized prompting template in Figure 1 and normalized via the same parsing procedure (JSON decode, key-order realignment to the canonical 58-attribute schema, and division by 5). These structured semantic

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System.
Rate the song on each MGPHot musical attribute as an integer 0-5 (0 =
absent/lowest, 5 = dominant/highest). Return only a JSON object with the
requested keys. No explanation.

User (anchor #1).
"Mrs. Robinson" by Simon & Garfunkel
Keys: ["Vocal Register", "Vocal Timbre Thin to Full", ..., "Focus on
Performance"]

Assistant (anchor #1).
{"Vocal Register": 4, "Vocal Timbre Thin to Full": 4, "Vocal Breathiness": 4,
..., "Focus on Performance": 4}

User (anchor #2).
"Master Of Puppets" by Metallica
Keys: ["Vocal Register", "Vocal Timbre Thin to Full", ..., "Focus on
Performance"]

Assistant (anchor #2).
{"Vocal Register": 4, "Vocal Timbre Thin to Full": 4, "Vocal Breathiness": 0,
..., "Focus on Performance": 4}

User (target).
"{title}" by {artist}
Keys: ["Vocal Register", "Vocal Timbre Thin to Full", ..., "Focus on
Performance"]

```

Figure 1: Prompt template used for MGPHot annotation (no_meta_few_shot variant). The system message fixes the 0–5 integer rating scale and JSON-only output format. Two multi-turn few-shot anchors – Simon & Garfunkel’s *Mrs. Robinson* and Metallica’s *Master of Puppets*, with ratings taken from the published MGPHot annotations – calibrate the model’s use of the 0–5 range. The Keys: list enumerates the canonical ordering of all 58 MGPHot attributes and forces the assistant to return them in the same order. Output integers are divided by 5 to obtain the [0, 1] features used downstream.

representations are later integrated with audio embeddings and behavioral signals for multimodal recommendation experiments.

3.3 LLM-Generated Features (Meta-Features)

Beyond the MGPHot schema, we extract a broader set of song-level meta-features using prompted LLM inference over lyrics and track metadata. These features target dimensions of meaning, structure, and cultural context that are not directly captured by audio-based encoders and are not captured by standard metadata fields such as genre tags or Spotify audio attributes. For each song, the LLM receives the artist name, track title, and available lyric text, and returns structured values for each feature via zero-shot prompting. We organize the extracted meta-features into five categories, summarized in Table 3:

Some of these attributes deliberately complement rather than duplicate existing signals. For instance, the profanity intensity feature provides a graded measure of explicit content, going beyond Spotify’s binary explicit flag. Temporal orientation (past/present/future framework) is orthogonal to valence and captures nostalgia or aspiration independently of positivity/negativity. The spoken word ratio distinguishes rap and poetry from sung melody in a way that is qualitatively different from Spotify’s *speechiness*, which conflates the two. Together, these meta-features extend the semantic reach of the benchmark by exposing lyric-level and cultural dimensions of songs. These properties are difficult to infer reliably from either audio embeddings or structured acoustic descriptors alone.

These features are particularly relevant for songs where lyrical semantics strongly influence listener preference. Their distributions are likely sparse for niche attributes but can provide valuable signals for personalization and cold-start recommendation.

3.4 Audio Features

We augment the song catalog with two sets of audio-derived features: handcrafted acoustic descriptors and dense neural audio embeddings. These representations complement semantic metadata by capturing sonic characteristics that cannot be inferred directly from lyrics or textual descriptions alone.

For the handcrafted descriptors, we compute a 63-dimensional feature vector using *Librosa* [21] for each song, comprising: 40 Mel-frequency cepstral coefficients (MFCCs, capturing timbre), 2 spectral centroid values (brightness), 14 spectral contrast values (harmonic texture), 2 spectral rolloff values (high-frequency energy distribution), 2 RMS energy values (loudness), 2 zero-crossing rate values (noise versus tonal signal distinction), and 1 tempo estimate (beats per minute). Additionally, structured audio attributes such as valence, energy, and danceability are retrieved for 47,113 songs (94.2% coverage) via the ReccoBeats API, used as a replacement for the now-deprecated Spotify audio features endpoint.

For richer acoustic representations, we embed each song using four pre-trained audio encoders: *CLAP* [36] (music-language contrastive pretraining), *MERT* [15] (music-specific self-supervised pretraining), *Music2Vec* [13] (contrastive self-supervised learning on music audio), and *EnCodec* [3] (neural audio codec representation). In addition to these neural encoders, we include MFCC-based representations as a classical handcrafted baseline. Audio files are retrieved through automated matching and download pipelines using track and artist metadata from the LastFM-1K catalog. Embeddings are computed offline and stored for downstream recommendation experiments.

In addition to audio representations, we generate lyric embeddings using multiple pretrained text encoders, including *MiniLM*, *BGE-M3*, *MPNet*, *Multilingual Sentence Encoder*, and *BERT*, using web-scraped lyrics aligned to the same song catalog.

Audio-derived features complement semantic metadata by modeling timbre, tempo, energy, instrumentation, and production characteristics that are not explicitly represented in text-based descriptions alone. The high feature coverage across the catalog enables large-scale multimodal recommendation experiments under both zero-shot and fine-tuned settings.

3.5 Benchmark Summary and Feature Coverage

Together, these data sources constitute a multimodal benchmark for session-based music recommendation. The interaction component, derived from LastFM-1K, provides sequential listening histories from 814 users across 7,073 listening sessions, enabling evaluation under session-aware recommendation settings. The content component enriches each song with multiple complementary feature representations, including structured semantic metadata, LLM-generated meta-features, handcrafted acoustic descriptors, neural audio embeddings, and lyric embeddings. Table 4 summarizes the coverage, diversity, and metadata statistics of the enriched Top-50k song catalog.

Table 3: LLM-generated lyric meta-features. Twelve structured fields plus four short free-text annotations are extracted per track by a three-model consensus (Llama-3.3-70B-Instruct, Qwen2.5-72B-Instruct, Mistral-Nemo-12B-Instruct) and validated via Fleiss’ κ / ICC(2,1) against heuristic baselines.

Category	Attributes (range)
Lyric structure	Rhyme scheme (AABB / ABAB / ABCB / free / irregular / mixed), internal rhyme density [0,1], rhyme complexity score (0–4, end-rhyme to dense polysyllabic chain), repetition ratio (hook / refrain density, [0,1]), vocabulary richness (1–5 ordinal type-token tier), profanity intensity (0–4 graded); free-text rhyme-pattern note.
Narrative perspective	Dominant perspective (1st / 2nd / 3rd / mixed), perspective shift across sections; free-text shift description.
Novelty & oddity	Title–content coherence [0,1], genre subversion score (0–4, judged against artist’s typical catalog), humor / irony presence (0–2 graded), spoken-word vs sung ratio [0,1]; free-text humor-type; subversion-rationale notes.

Table 4: Top-50k song catalog statistics.

General		Coverage / Distribution	
Metric	Value	Metric	Value
Total songs	50,029	Audio features	47,113 (94.2%)
Unique artists	5,751	Preview URL	32,142 (64.2%)
Avg. play count	49.7	Genre tag	28,105 (56.2%)
Median play count	31.0	Unique genres	81
Avg. duration (min)	4.14	English	36,373 (72.7%)
Top genre: Rock	10,774	Non-English	13,536 (27.1%)
Electronic	3,463	Unknown language	120 (0.2%)
Pop	1,441	Unique language codes	42
Metal	1,368		

The resulting benchmark combines collaborative interaction signals with semantic and acoustic representations, allowing controlled evaluation of how different modalities contribute to recommendation quality. In particular, the benchmark supports analysis of multimodal fusion strategies, semantic grounding, and content-aware recommendation under both zero-shot and fine-tuned settings.

A notable characteristic of the benchmark is its diversity in both language and genre coverage. Approximately 27.1% of songs are non-English tracks spanning 42 language codes, while genre annotations cover 81 unique genres. These properties make the dataset suitable for studying recommendation robustness across heterogeneous musical content and multilingual recommendation scenarios.

Overall, the benchmark is designed to support systematic investigation of multimodal sequential recommendation, particularly in sparse and cold-start settings where interaction-only representations may be insufficient. In the next section, we describe how these multimodal representations are integrated into sequential recommendation architectures.

4 Evaluation Method

We propose a multimodal sequential music recommendation framework that addresses two key limitations of existing LLM-based recommenders. First, prior methods represent items solely through learned ID embeddings, which contain limited information about a

song’s acoustic properties, semantic content, or lyrical characteristics. Second, they treat all interactions in a user’s listening history as equally positive signals, ignoring behavioral evidence derived from how long a user actually listened to a track.

Our framework addresses both limitations through two coupled components. On the item side, we construct content-aware item representations by combining three complementary modalities (Figure 2): acoustic embeddings that capture the sonic characteristics of a song, lyric embeddings that encode semantic and emotional content, and structured semantic metadata that provides high-level musicological attributes. These multimodal representations are projected into the LLM embedding space, enabling the model to reason over song content rather than relying exclusively on item identifiers.

On the interaction side, we incorporate a completion ratio signal derived from listening duration relative to track length. A user who listens to nearly an entire song expresses a qualitatively different preference signal than one who skips after a few seconds, yet conventional recommendation frameworks typically treat both interactions identically. By incorporating completion ratio information, the framework distinguishes strong engagement from incidental exposure during sequential recommendation.

Figure 3 illustrates the overall architecture of the proposed framework, including the integration of multimodal item representations and completion ratio signals into the LLM-based recommendation pipeline.

4.1 Semantic Metadata

Beyond raw audio and lyrical signals, we incorporate high-level musicological features to provide an interpretable semantic layer. We adopt the MGPHot annotation schema [23], which defines 58 musical attributes across seven categories: Lyrics (mood dimensions such as Angry, Sad, Happy, Humorous, Love), Sonority (Live Recording, Acoustic, Electric), Composition (Focus on Lead Vocal, Melody, Lyrics), Vocals (Register, Timbre, Breathiness, Smoothness), Harmony (Minor/Major Key Tonality, Harmonic Sophistication), Rhythm (Tempo, Swing, Syncopation, Danceability), and Instrumentation (Drum Set, Electric Guitar, Acoustic Guitar, Piano). Each attribute is represented on a normalized scale from 0 to 1.

As an intermediate metadata configuration, we also incorporate acoustic attributes retrieved via the ReccoBeats / Spotify Extended

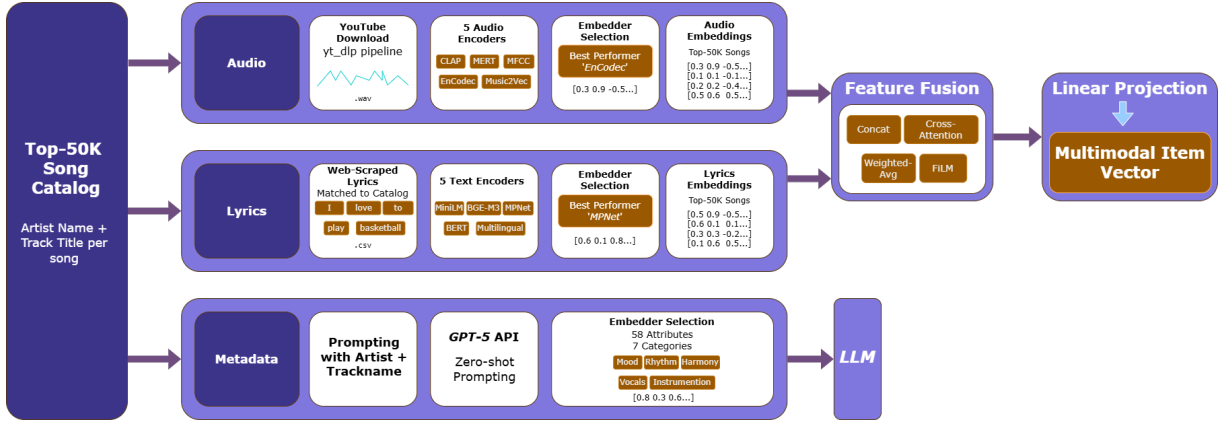


Figure 2: Multimodal feature extraction pipeline.

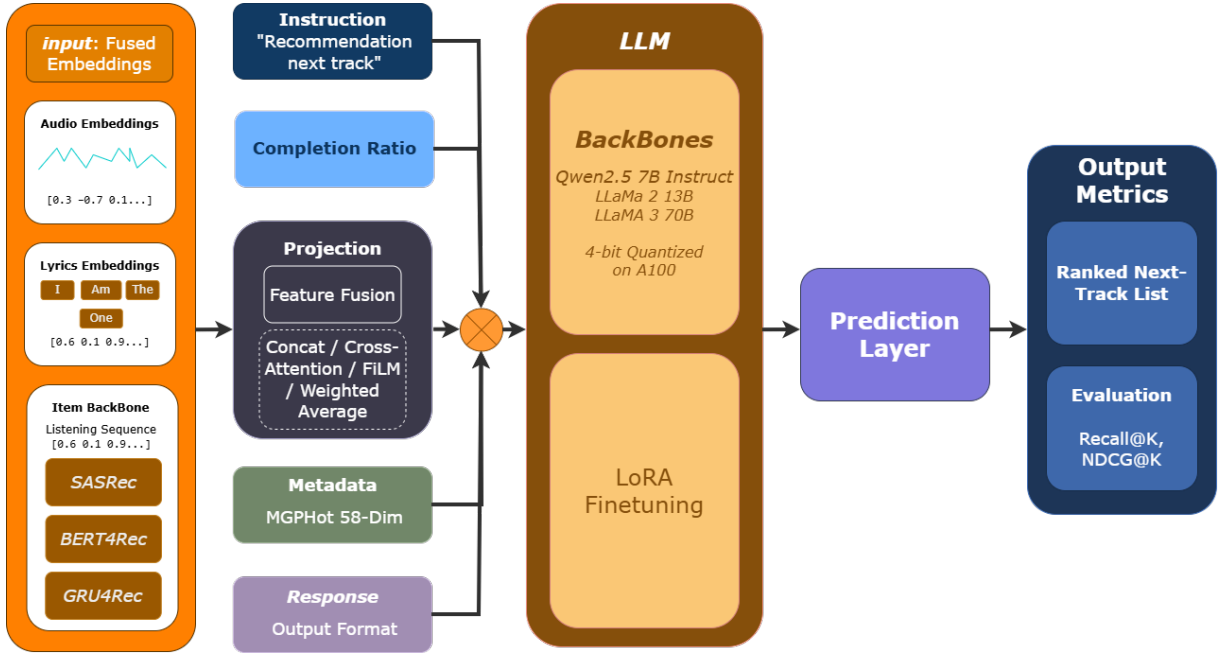


Figure 3: Overview of the proposed multimodal recommendation architecture.

Audio APIs (via RapidAPI) [27] as a proxy for the deprecated Spotify Audio Features API. These structured features (e.g., valence, energy, danceability, and instrumentality) are compact, acoustically grounded descriptors, and allow us to evaluate the benefit of structured metadata independently of the MGPHot extraction pipeline.

4.2 Multimodal Fusion Strategies

A core design question in our framework is how to combine item ID embeddings with content modality embeddings, specifically audio and lyric embeddings, before passing them into the LLM. Rather

than adopting a single fusion strategy, we treat fusion as an empirical design choice and evaluate four approaches with increasing representational capacity.

Concatenation is the simplest baseline: the item ID embedding and content embeddings are concatenated along the feature dimension and passed through a shared linear projection. This approach makes no assumptions about the relationship between modalities and lets the projection layer learn how to weight them.

Weighted Sum linearly combines the item ID and content embeddings using learned scalar weights, producing a single vector of the same dimensionality as each input. This is more parameter-efficient than concatenation and explicitly models the relative importance of each modality, but assumes a linear relationship between them.

Cross-Attention treats the item ID embedding as a query and the content embeddings as keys and values, allowing the model to selectively attend to the most relevant parts of the audio or lyric representation conditioned on the item identity.

FiLM (Feature-wise Linear Modulation) [24] conditions the item ID embedding on the content embeddings via affine transformations; the content embeddings predict scaling and shifting parameters applied element-wise. This allows content signals to modulate the collaborative filtering representation rather than simply being added to it, potentially preserving the structure of the pretrained collaborative filtering space.

We evaluate all four strategies across audio and lyric modalities independently, then apply the best-performing strategy to the combined audio and lyric setting. For semantic metadata, rather than performing embedding-level fusion, we incorporate it directly into the LLM’s instruction prompt, allowing the model to reason over structured musical attributes in natural language. For the completion ratio signal, we explore two incorporation strategies: prompt injection, where the completion ratio is expressed as a natural language signal in the instruction prompt, and concatenation with existing item embeddings. We evaluate both approaches to determine whether engagement signals are more effectively utilized in textual or embedding representations.

5 Experimental Setup

5.1 Datasets

Data Source. We used the **LastFM-1K** dataset, a publicly available collection of listening histories for approximately 1,000 users of the Last.fm music platform, including artist name, track name, and play counts. Since the dataset provides only raw listening events without structured song representations, we construct an enriched version through a multi-stage pipeline. To support multimodal recommendation experiments, we enrich the dataset with structured semantic metadata, LLM-generated musicological attributes, audio embeddings, and lyric embeddings.

Preprocessing. We apply the following preprocessing steps to construct the session-based recommendation dataset. First, we filter out users with fewer than 1,000 total listening events, and tracks with fewer than 7 plays across the dataset, to ensure sufficient interaction density for sequential modeling. We then segment each user’s listening history into sessions by splitting on gaps of more than 20 minutes of inactivity, which reflects natural listening behavior boundaries. Within each session, duplicate tracks are removed. Sessions with fewer than 10 songs are discarded. These filtering and segmentation steps result in a dataset of high-quality, contextually coherent listening sessions.

Data Splits. The LastFM-1K dataset provides no off-the-shelf splits. We first segment listening histories into sessions following Tran et al. [30], then define train/validation/test splits using the global temporal cutoff methodology of Abbattista et al. [1]. The 90th-percentile timestamp serves as the test cutoff, placing the most recent 10% of interactions in the test set. For validation, ~10% of users (~81, sampled from the 814-user cohort) are randomly selected and an analogous 90th-percentile temporal cutoff is applied

within their pre-test history; 60 of these end up with a non-empty validation window. The final train / val / test cohorts contain 638 / 60 / 638 users respectively. The same fixed splits are applied to all models to ensure fair comparison, with the validation set used exclusively for hyperparameter tuning and the test set reserved solely for final evaluation and is never used during training or development.

The full data enrichment pipeline, illustrating how raw LastFM-1K listening events are transformed into a richly annotated multimodal dataset, is shown in Figure 4.

5.2 Evaluation

We evaluate all model variants on a fixed held-out test set across four experimental settings: (1) ID-only baselines, (2) semantic metadata augmentation, (3) audio embedding augmentation, and (4) engagement-aware modeling. This staged evaluation setup enables controlled analysis of the contribution of each enrichment component to recommendation performance.

We evaluate recommendation quality using standard top-k ranking metrics computed over the test set. For each test session, the model ranks all candidate items and the held-out ground-truth item is evaluated within the ranked candidate list.

We adopt Recall@K and Normalized Discounted Cumulative Gain (NDCG@K) at $K \in \{5, 10, 20\}$ as our primary metrics, consistent with the evaluation practice of our baselines, which enables direct comparison. Recall@K measures whether the ground-truth item appears within the top-K recommendations, rewarding any correct retrieval regardless of rank. NDCG@K is a position-sensitive metric that assigns higher scores to correct items ranked closer to the top of the list, providing a more nuanced measure of ranking quality.

We evaluate all models under the same fixed test set and metric suite to ensure comparability. We do not report metrics at $k = 100$ in primary results, as at large cutoffs even weak models appear competitive and the metric loses discriminative power; however, we include $k = 100$ results in supplementary analysis for completeness.

5.3 Baselines

We benchmark our approach against three recent methods, re-training and re-evaluating all baselines on our dataset partitions to ensure a consistent experimental setting. Where available, we use official open-source implementations and released checkpoints.

E4SRec [12] is our primary baseline, which integrates LLMs with sequential recommendation by injecting pretrained item ID embeddings into the LLM input space. Since it operates solely on item identifiers without multimodal enrichment, it serves as a direct reference point for evaluating the gains introduced by multimodal enrichment.

LLaRA [17] addresses the semantic gap between item ID embeddings and LLM input spaces via a hybrid prompting strategy that combines ID-based behavioral signals with textual item descriptions. This baseline isolates the contribution of our multimodal enrichment over existing ID-to-text alignment approaches.

LC-Rec [44] bridges collaborative and language representations through hierarchical semantic IDs generated from vector-quantized

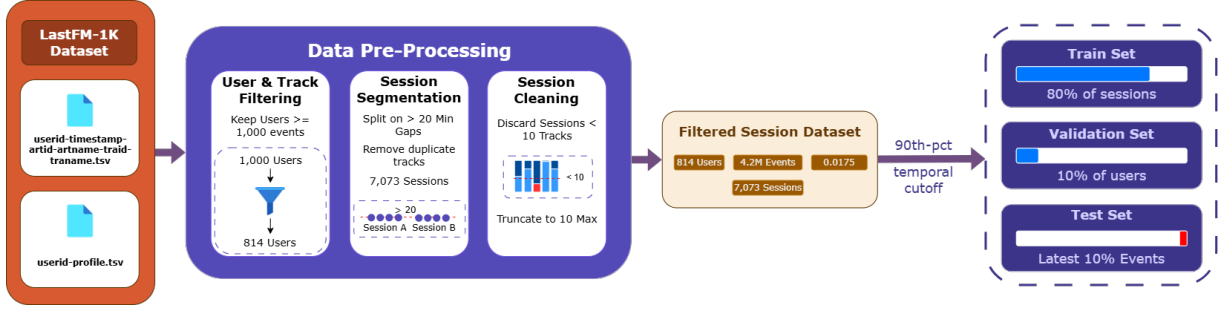


Figure 4: Data Pre-processing pipeline.

item text embeddings. This enables the LLM to reason over structured item relationships alongside interaction sequences.

Together, these baselines span LLM-based sequential recommendation (E4SRec), ID-to-text alignment (LLaRA), and collaborative semantic integration (LC-Rec), enabling comprehensive evaluation across multiple recommendation paradigms. To ensure fair comparison, all baselines are re-trained and evaluated on our preprocessed LastFM-1K splits using the same evaluation protocol and metrics.

5.4 Implementation Details

Our framework is implemented using PyTorch for model training and inference, HuggingFace Transformers and PEFT for LLM loading and LoRA fine-tuning, and NumPy and Pandas for data preprocessing and feature management.

Pre-trained audio encoders, including CLAP, MERT, Music2Vec, and EnCodec, are loaded from HuggingFace Hub. Text encoders such as MiniLM, BGE-M3, MPNet, multilingual sentence encoders, and BERT are used for lyric representation learning.

The evaluated LLM backbones include Llama-2-7B, Qwen2.5-7B-Instruct, Qwen2.5-32B, Qwen2.5-32B-Instruct, and Llama-Song-Stream-3B-Instruct. Larger models are loaded with 4-bit quantization when necessary to fit within available GPU memory. Audio retrieval and download pipelines use yt-dlp. Semantic metadata extraction is handled via the OpenAI API (GPT-4o). Structured audio metadata is retrieved using the ReccoBeats/SpotifyEA API.

Zero-Shot Configuration. In zero-shot experiments, LLM weights are frozen and no task-specific training is performed. Item representations are constructed from the multimodal features described in Section 3, combined using the fusion strategies described in Section 4 and projected into the LLM hidden space.

Fine-Tuning Configuration. Fine-tuning experiments use LoRA-based parameter-efficient adaptation applied to the LLM backbone. Fine-tuning experiments use the same backbone & modality configuration grid as zero-shot experiments to enable direct comparison.

Experiments are conducted on the Unity Cluster using NVIDIA A100 GPUs for multimodal embedding extraction, zero-shot inference, and LoRA fine-tuning. Audio files, metadata, and pretrained embeddings are stored offline and reused across experiments to support efficient large-scale evaluation.

5.5 Zero-Shot Results and Analysis

We evaluate all backbone and modality configurations under zero-shot conditions, where LLM weights remain frozen and no task-specific training is performed. This establishes the degree to which pre-trained LLMs can leverage multimodal content signals for sequential music recommendation without any fine-tuning.

Embedder Selection. Before running the full backbone comparison, we conduct a preliminary embedder selection study using Qwen2.5-7B-Instruct to identify the strongest audio and lyric embedders respectively. Based on these results, we select EnCodec as the primary audio embedder due to its strong performance across evaluation metrics and cutoffs, and MPNet as the primary lyric embedder due to its most consistent ranking performance (NDCG, MRR) across cutoffs. All subsequent experiments use these two embedders.

Backbone and Modality Comparison. Table 5 reports zero-shot Recall and NDCG at $k \in \{5, 10, 20\}$ across three LLM backbones and six input configurations.

Analysis.

Effect of Modality: We analyze the impact of incorporating additional modalities, namely audio embeddings, lyric embeddings, and metadata, on recommendation performance across different backbones.

Overall, lyric embeddings emerge as the most effective modality, consistently improving both recall and ranking quality. For instance, in the case of Qwen2.5-7B-Instruct, incorporating lyric embeddings yields substantial gains across all metrics, improving Recall@5 from 0.024 to 0.050 and NDCG@20 from 0.032 to 0.061. Similar trends are observed for LLaMA-7B, where lyric-based augmentation leads to the best NDCG scores across all cutoffs. This suggests that lyric semantics align naturally with the language modeling capabilities of LLM-based recommenders.

In contrast, audio embeddings produce mixed results across backbones. While certain configurations benefit from audio features (e.g., Qwen2.5-7B-Instruct shows improvements at higher cutoffs), other models such as LLaMA-Song-Stream-3B-Instruct experience noticeable degradation when audio embeddings are introduced. This indicates that audio representations may be harder to integrate effectively, potentially due to weaker alignment between acoustic representations and the LLM embedding space.

Table 5: Performance comparison across LLM backbones and input configurations.

LLM Backbone	Configuration	@5		@10		@20	
		Recall	NDCG	Recall	NDCG	Recall	NDCG
Qwen2.5-7B-Instruct	SASRec only	0.017	0.011	0.044	0.019	0.083	0.029
	SASRec + Audio + Lyric + Metadata	0.025	0.014	0.049	0.022	0.103	0.035
	BERT4Rec only	0.027	0.016	0.047	0.022	0.099	0.035
	BERT4Rec + Audio + Lyric + Metadata	0.038	0.024	0.067	0.033	0.114	0.045
	GRU4Rec only	0.034	0.020	0.056	0.027	0.091	0.035
	GRU4Rec + Audio + Lyric + Metadata	0.025	0.015	0.060	0.026	0.107	0.038
Llama-2-13B	SASRec only	0.027	0.013	0.056	0.022	0.100	0.033
	SASRec + Audio + Lyric + Metadata	0.024	0.014	0.044	0.020	0.088	0.031
	BERT4Rec only	0.017	0.009	0.055	0.021	0.103	0.033
	BERT4Rec + Audio + Lyric + Metadata	0.022	0.014	0.053	0.024	0.107	0.037
	GRU4Rec only	0.027	0.015	0.050	0.023	0.099	0.035
	GRU4Rec + Audio + Lyric + Metadata	0.027	0.016	0.044	0.022	0.085	0.032
Llama-3-70B	SASRec only	0.019	0.011	0.039	0.018	0.100	0.033
	SASRec + Audio + Lyric + Metadata	0.028	0.019	0.055	0.028	0.092	0.037
	BERT4Rec only	0.034	0.023	0.063	0.032	0.105	0.042
	BERT4Rec + Audio + Lyric + Metadata	0.020	0.013	0.033	0.017	0.061	0.024
	GRU4Rec only	0.019	0.012	0.052	0.023	0.102	0.035
	GRU4Rec + Audio + Lyric + Metadata	0.022	0.015	0.045	0.022	0.105	0.037

The contribution of metadata is relatively modest and less consistent across models. While minor improvements are observed in select cases (e.g., LLaMA-7B at Recall@5), metadata generally underperforms compared to lyric embeddings. This suggests that lightweight structured metadata alone may be insufficient to capture the richer semantic relationships leveraged by LLM-based recommenders.

Finally, we observe that combining multiple modalities does not necessarily lead to additive gains. In particular, for Qwen2.5-7B-Instruct, the full multimodal configuration performs worse than using lyric embeddings alone. These findings emphasize the importance of robust multimodal fusion mechanisms.

Effect of Backbone: We further examine how the choice of LLM backbone influences recommendation performance and its ability to leverage multimodal signals.

Among all models, Qwen2.5-7B-Instruct achieves the strongest overall performance, particularly when augmented with lyric embeddings. It not only attains the highest scores across multiple metrics but also demonstrates significant improvements when additional modalities are introduced. This indicates that instruction-tuned Qwen models are better equipped to utilize auxiliary information for recommendation tasks.

In comparison, LLaMA-7B exhibits moderate improvements with modality augmentation, but the gains are less pronounced than those observed for Qwen models. This suggests relatively weaker alignment between LLaMA representations and the recommendation task.

Interestingly, LLaMA-Song-Stream-3B-Instruct underperforms despite being domain-specific. The baseline SASRec configuration consistently outperforms multimodal variants, indicating that the model struggles to effectively incorporate additional modalities.

This may stem from limited model capacity or ineffective integration of external embeddings.

Scaling to larger models, such as Qwen2.5-32B, leads to improvements in higher-cutoff recall (e.g., Recall@20), suggesting better coverage of relevant items. However, gains in NDCG are less consistent, suggesting that increased model scale does not uniformly translate to improved ranking quality.

Finally, comparing base and instruction-tuned variants of Qwen2.5-32B reveals that instruction tuning appears to improve the model’s ability to utilize structured metadata inputs, as evidenced by improved performance at higher cutoffs.

6 Ablation Study

We conduct ablation studies to analyze the contribution of different embedding models, multimodal fusion strategies, sequential backbones, and completion-ratio incorporation methods. All ablations are evaluated under controlled settings using the same train/validation/test splits and evaluation protocol.

6.1 Acoustic Embeddings

We evaluate the usefulness of acoustic embeddings over five audio encoders (*CLAP* [36], *MERT* [15], *Music2Vec* [13], *EnCodec* [3] and MFCC) independently under zero-shot inference with Qwen2.5-7B-Instruct. Table 6 reports Recall, NDCG, MRR, and Precision at $K \in \{5, 10, 20\}$.

EnCodec achieves the strongest overall performance across most metrics and cutoffs, consistently outperforming the SASRec baseline, so we select EnCodec as the primary audio embedder for the main experiments. We additionally evaluate embedding fusion strategies in subsequent ablation experiments.

Table 6: Audio embeddings performance.

K	Model	Recall	NDCG	MRR	Precision
5	SASRec	0.024	0.014	0.011	0.005
	CLAP	0.022	0.015	0.012	0.004
	MERT	0.033	0.022	0.019	0.007
	MUSIC2VEC	0.025	0.015	0.012	0.005
	ENCODEC	0.034	0.024	0.020	0.007
	MFCC	0.017	0.008	0.005	0.003
10	SASRec	0.039	0.019	0.013	0.004
	CLAP	0.039	0.020	0.014	0.004
	MERT	0.058	0.030	0.022	0.006
	MUSIC2VEC	0.044	0.021	0.014	0.004
	ENCODEC	0.072	0.035	0.025	0.007
	MFCC	0.047	0.018	0.009	0.005
20	SASRec	0.089	0.032	0.016	0.004
	CLAP	0.085	0.031	0.017	0.004
	MERT	0.108	0.042	0.025	0.005
	MUSIC2VEC	0.086	0.032	0.017	0.004
	ENCODEC	0.121	0.048	0.028	0.006
	MFCC	0.086	0.027	0.012	0.004

Table 7: Lyric embeddings performance.

K	Model	Recall	NDCG	MRR	Precision
5	SASRec	0.024	0.014	0.011	0.005
	MINILM	0.050	0.019	0.010	0.010
	BGEM3	0.000	0.000	0.000	0.000
	MPNET	0.050	0.050	0.050	0.010
	MULTILINGUAL	0.050	0.032	0.025	0.010
	BERT	0.000	0.000	0.000	0.000
10	SASRec	0.039	0.019	0.013	0.004
	MINILM	0.100	0.035	0.016	0.010
	BGEM3	0.000	0.000	0.000	0.000
	MPNET	0.050	0.050	0.050	0.005
	MULTILINGUAL	0.050	0.032	0.025	0.005
	BERT	0.000	0.000	0.000	0.000
20	SASRec	0.089	0.032	0.016	0.004
	MINILM	0.100	0.035	0.016	0.005
	BGEM3	0.150	0.037	0.010	0.008
	MPNET	0.100	0.061	0.053	0.005
	MULTILINGUAL	0.050	0.032	0.025	0.002
	BERT	0.050	0.011	0.002	0.002

6.2 Lyric Embeddings

Similar to audio embedders, we evaluate all five lyric embedders (*MiniLM*, *BGEM3*, *MPNet*, a multilingual sentence transformer, and *BERT*) independently under zero-shot inference with Qwen2.5-7B-Instruct. Table 7 reports Recall, NDCG, MRR, and Precision at $k \in \{5, 10, 20\}$.

Among the five lyric embedders, BERT and BGEM3 fail to produce meaningful recommendations at tight cutoffs ($k \leq 10$), scoring near zero across all metrics, suggesting weaker alignment between these representations and the recommendation objective. MPNet achieves the strongest overall ranking quality, achieving the best NDCG and MRR across all cutoffs making it the most reliable lyric

embedder for recommendation despite not always achieving the highest Recall. MiniLM achieves high Recall at $k = 10$ but lower ranking precision, and BGEM3 improves at larger cutoffs primarily through broader retrieval coverage rather than precise ranking. We select MPNet as the primary lyric embedder for subsequent experiments based on its consistently superior ranking quality.

6.3 Metadata

We study the contribution of semantic metadata under zero-shot inference across three LLM backbones (Qwen2.5-7B, Llama-2-7B, Llama-3-70B), three item-ID backbones (SASRec, BERT4Rec, GRU4Rec), and ten metadata configurations that vary the source of metadata (Spotify-derived, MGPHot-derived, or their combination), the level of feature curation (all features, the “All with signal” subset, or the “Best with signal” subset), and whether completion-ratio (CR) signals are incorporated. The two curated subsets are themselves the output of a separate greedy feature-selection search run on LLaMA-2-13B + top-50k LastFM, which only manipulates the prompt’s feature/annotation columns and re-evaluates NDCG@10. The search first performs a single-column ablation over all 76 Spotify + MGPHot columns and retains the 15 that individually beat $0.95\times$ the no-metadata baseline – we call this set ALL WITH SIGNAL ($|S| = 15$: Abstract Lyrics, Angry Lyrics, Cut Time Feel, Danceability, Focus on Form, Odd Meter, String Ensemble, Swing Feel, Synthesizer, Vocal Breathiness, Vocal Register, duration_ms, key, language, loudness). It then runs greedy forward selection followed by bounded local search over those 15 columns and converges to the 2-column subset $S^* = \{\text{key, Synthesizer}\}$, which lifts NDCG@10 from 0.021 to 0.025 (+19%); we call this BEST WITH SIGNAL. The 90-cell sweep below transplants these two subsets (and their union/superset counterparts) into a wider grid of LLMs and item-ID backbones. Table 13 reports per-LLM aggregates across all 30 (item-ID backbone $\times$ config) cells, Table 14 reports the best metadata configuration per (LLM, item-ID backbone) cell, and Tables 15, 16, and 17 report NDCG@10, Recall@20, and MRR@10 for all 90 cells.

Prompt curation outweighs raw parameter count. The single strongest cell in the entire sweep is **Llama-2-7B + BERT4Rec + All-with-signal + CR**, reaching NDCG@10 = 0.028, Recall@10 = 0.066, and Recall@20 = 0.107 – a +56% relative improvement over the global mean NDCG@10 of 0.018. Notably, this peak is attained by a 7B model, not the 70B one, suggesting that prompt curation outweighs raw parameter count in this zero-shot, frozen-embedding regime.

LLM scale is not the dominant lever. All three LLMs converge to nearly identical aggregate behavior (mean NDCG@10 = 0.018 ± 0.003 for each; Table 13). The 7B→70B step from Llama-2 to Llama-3 yields no measurable mean improvement, and per-cell winner counts are roughly balanced across the 30 (backbone $\times$ config) cells per model (Qwen2.5-7B wins 8, Llama-2-7B wins 12, Llama-3-70B wins 10). On a frozen-embeddings zero-shot setup, additional LLM scale does not translate to better ranking quality.

Item-ID backbone is a secondary effect. SASRec achieves the highest mean NDCG@10 (0.019) across all 30 (LLM $\times$ config) cells per backbone, with BERT4Rec and GRU4Rec close behind (both

Table 8: Zero-shot results for Qwen2.5-7B-Instruct.

LLM Backbone	Item Backbone	Configuration	@5		@10		@20	
			Recall	NDCG	Recall	NDCG	Recall	NDCG
Qwen2.5-7B-Instruct	SASRec	ItemID only	0.033	0.020	0.056	0.027	0.100	0.038
		+ Audio (Concat)	0.016	0.009	0.041	0.017	0.097	0.031
		+ Audio (Weighted Avg.)	0.020	0.013	0.049	0.022	0.100	0.034
		+ Audio (Cross-Attn.)	0.038	0.019	0.069	0.029	0.110	0.040
		+ Audio (FiLM)	0.024	0.013	0.044	0.019	0.094	0.032
		+ Lyric (Concat)	0.022	0.013	0.039	0.018	0.089	0.031
		+ Lyric (Weighted Avg.)	0.034	0.019	0.063	0.028	0.094	0.035
		+ Lyric (Cross-Attn.)	0.028	0.016	0.049	0.023	0.088	0.033
		+ Lyric (FiLM)	0.028	0.016	0.050	0.023	0.100	0.036
		+ Audio + Lyric (Concat)	0.017	0.009	0.044	0.017	0.091	0.029
		+ Audio + Lyric(Concat) + CR(Prompt)	0.025	0.016	0.042	0.022	0.097	0.035
		+ Audio + Lyric(Concat) + CR(Embed)	0.017	0.009	0.036	0.015	0.097	0.035
		+ Audio + Lyric (Weighted Avg.)	0.020	0.014	0.039	0.020	0.091	0.029
		+ Audio + Lyric (Cross-Attn.)	0.038	0.024	0.045	0.026	0.102	0.040
		+ Audio + Lyric (FiLM)	0.024	0.015	0.053	0.024	0.099	0.036
	BERT4Rec	ItemID only	0.033	0.020	0.055	0.026	0.102	0.038
		+ Audio (Concat)	0.041	0.025	0.056	0.030	0.103	0.042
		+ Audio (Weighted Avg.)	0.024	0.015	0.041	0.021	0.100	0.036
		+ Audio (Cross-Attn.)	0.033	0.019	0.056	0.026	0.103	0.038
		+ Audio (FiLM)	0.024	0.014	0.047	0.021	0.114	0.038
		+ Lyric (Concat)	0.024	0.017	0.050	0.026	0.110	0.041
		+ Lyric (Weighted Avg.)	0.027	0.015	0.047	0.022	0.102	0.035
		+ Lyric (Cross-Attn.)	0.028	0.015	0.053	0.023	0.089	0.032
		+ Lyric (FiLM)	0.024	0.013	0.050	0.022	0.092	0.032
		+ Audio + Lyric (Concat)	0.025	0.013	0.052	0.021	0.103	0.034
		+ Audio + Lyric (Weighted Avg.)	0.022	0.011	0.052	0.021	0.097	0.032
		+ Audio + Lyric (Cross-Attn.)	0.014	0.008	0.041	0.017	0.088	0.029
		+ Audio + Lyric (FiLM)	0.014	0.010	0.047	0.020	0.100	0.034
	GRU4Rec	ItemID only	0.025	0.015	0.058	0.026	0.108	0.038
		+ Audio (Concat)	0.025	0.013	0.044	0.018	0.100	0.033
		+ Audio (Weighted Avg.)	0.025	0.013	0.049	0.021	0.113	0.037
		+ Audio (Cross-Attn.)	0.027	0.015	0.050	0.022	0.103	0.035
		+ Audio (FiLM)	0.025	0.014	0.045	0.020	0.094	0.032
		+ Lyric (Concat)	0.028	0.018	0.058	0.028	0.100	0.038
		+ Lyric (Weighted Avg.)	0.022	0.013	0.053	0.023	0.103	0.035
		+ Lyric (Cross-Attn.)	0.028	0.016	0.061	0.027	0.099	0.036
		+ Lyric (FiLM)	0.025	0.014	0.045	0.021	0.096	0.033
		+ Audio + Lyric (Concat)	0.036	0.021	0.066	0.031	0.114	0.043
		+ Audio + Lyric (Weighted Avg.)	0.024	0.013	0.047	0.021	0.092	0.032
		+ Audio + Lyric (Cross-Attn.)	0.017	0.008	0.041	0.016	0.091	0.028
		+ Audio + Lyric (FiLM)	0.025	0.014	0.047	0.021	0.107	0.035

0.018). The max–min spread of mean NDCG@10 across backbones is only 1.1×10^{-3} . BERT4Rec, however, produces the single highest cell (0.028), indicating higher variance: SASRec is the steadier choice, BERT4Rec the more upside-bound one. Once the LLM front-end is in the loop, the choice of item-ID backbone is far less consequential than the choice of metadata configuration.

Metadata curation dominates. The strongest config on average is ALL-WITH-SIGNAL + CR (mean NDCG@10 = 0.020); the weakest is

ALL COMBINED (0.015). The full 76-column ALL COMBINED prompt actively hurts ranking quality across every metric and every cutoff – adding more metadata is not free, because extra columns inject noise that the LLM must filter through. The curated 15-column ALL-WITH-SIGNAL subset and the minimal 2-column BEST-WITH-SIGNAL subset are competitive with or superior to the full feature set on most cells, supporting the broader hypothesis that prompt parsimony pays off for LLM-based recommenders.

Table 9: Fine-tuned results for Qwen2.5-7B-Instruct (SASRec backbone).

LLM Backbone	Item Backbone	Configuration	@5		@10		@20	
			Recall	NDCG	Recall	NDCG	Recall	NDCG
Qwen2.5 7B-Instruct Finetuned	SASRec	ItemID only	0.188	0.102	0.268	0.128	0.346	0.148
		+ Audio (Concat)	0.226	0.123	0.326	0.155	0.398	0.173
		+ Audio (Weighted Avg.)	0.210	0.115	0.313	0.149	0.386	0.167
		+ Audio (Cross-Attn.)	0.224	0.121	0.323	0.153	0.400	0.173
		+ Audio (FiLM)	0.221	0.118	0.317	0.150	0.392	0.169
		+ Lyric (Concat)	0.227	0.123	0.324	0.155	0.390	0.172
		+ Lyric (Weighted Avg.)	0.212	0.115	0.288	0.140	0.389	0.166
		+ Lyric (Cross-Attn.)	0.216	0.116	0.293	0.140	0.381	0.162
		+ Lyric (FiLM)	0.218	0.118	0.304	0.146	0.403	0.171
		+ Audio + Lyric (Concat)	0.229	0.124	0.313	0.152	0.398	0.173
		+ Audio + Lyric (Weighted Avg.)	0.224	0.120	0.306	0.146	0.400	0.170
		+ Audio + Lyric (Cross-Attn.)	0.224	0.122	0.318	0.153	0.398	0.172
		+ Audio + Lyric (FiLM)	0.216	0.121	0.317	0.153	0.384	0.170

LLM-generated annotations vs. Spotify features vs. no meta-data. A direct head-to-head between sources isolates where the signal lives. The E4SRec baseline reaches $\text{NDCG}@10 = 0.021$; injecting all 18 Spotify columns drops it to 0.019 (−9%), injecting all 58 MGPHot perceptual annotations leaves it essentially unchanged at 0.021 (−1%), and naively concatenating both into a 76-column prompt drops it further to 0.015 (−29%). The same pattern holds in the 90-cell sweep: mean $\text{NDCG}@10$ is 0.018 for both ALL SPOTIFY and ALL MGPHOT, but only 0.015 for ALL COMBINED; on Recall@20, ALL MGPHOT (mean 0.099) additionally edges out ALL SPOTIFY (mean 0.090). Two facts follow. First, the LLM-generated perceptual annotations preserve baseline ranking quality and improve retrieval, whereas raw Spotify numerics on their own actively hurt – consistent with the hypothesis that LLMs can parse descriptive labels (e.g., “Cut Time Feel”, “String Ensemble”) but struggle to extract usable signal from bare numerics (e.g., $\text{tempo}=120$, $\text{energy}=0.65$). Second, mixing the two sources naively is strictly worse than either alone – the only configurations that beat the no-metadata baseline are the curated 15-column ALL-WITH-SIGNAL subset (mean $\text{NDCG}@10 = 0.020$, greedy-search peak 0.025) and the 2-column BEST-WITH-SIGNAL subset (0.025). ALL-WITH-SIGNAL is itself 11/15 MGPHot annotations + 4/15 Spotify numerics, and BEST-WITH-SIGNAL = {key, Synthesizer} contains exactly one column from each modality. The practical takeaway is that LLM-generated metadata is the dominant source of useful signal, but a small Spotify residual is still complementary; injecting either source uncured – and especially their union – adds more noise than information.

Completion-ratio is a tunable knob, not a free win. Across the 45 matched (LLM × backbone × base config) pairs, adding the completion-ratio signal shifts $\text{NDCG}@10$ by only $+2 \times 10^{-4}$ on average – essentially zero net effect. CR helps in 22 pairs, hurts in 20, and is neutral in 3. Per-LLM and per-backbone mean deltas are all within $\pm 1.4 \times 10^{-3}$, with BERT4Rec showing the most consistent

benefit ($+1.4 \times 10^{-3}$, positive in 9/15 pairs) and SASRec the least (-1.1×10^{-3} , positive in 7/15). The single largest gain (+0.007 on Llama-2-7B + BERT4Rec + ALL-WITH-SIGNAL) and the single largest loss (−0.010 on Qwen2.5-7B + SASRec + ALL MGPHOT) are of similar magnitude, so CR should be treated as a hyperparameter to be searched per (model, backbone, config) recipe rather than a default-on enhancement.

No cross-LLM agreement on the best config. For every item-ID backbone, the three LLMs each pick a different best-performing metadata configuration (Table 14): on SASRec the winners are ALL-WITH-SIGNAL + CR, ALL SPOTIFY + CR, and ALL MGPHOT; on BERT4Rec they are BEST-WITH-SIGNAL + CR, ALL-WITH-SIGNAL + CR, and ALL COMBINED + CR; on GRU4Rec they are ALL SPOTIFY + CR, ALL-WITH-SIGNAL, and ALL-WITH-SIGNAL + CR. This indicates that the optimal prompt depends jointly on the LLM and the item-ID backbone, not just on the metadata itself – there is no universally best config.

Retrieval vs. ranking: configs specialize. Cross-metric analysis reveals a consistent dichotomy. ALL-WITH-SIGNAL + CR wins on $\text{NDCG}@10$, $\text{NDCG}@20$, $\text{MRR}@10$, and $\text{Recall}@10$ (top-of-list precision), while ALL MGPHOT wins on $\text{Recall}@20$ (broader retrieval). $\text{NDCG}@10$ and $\text{MRR}@10$ are highly rank-correlated across cells (Kendall $\tau = +0.87$), but the agreement between $\text{NDCG}@10$ and $\text{Recall}@K$ weakens as K grows ($\tau = +0.88$ at $K=10$, $+0.48$ at $K=20$, near zero at $K=100$). At $\text{Recall}@100$ all configs converge to ≈ 0.49 – 0.51 , meaning the LLM-prompt differences live almost entirely in the top of the list. The practical takeaway is that compact curated subsets are best for downstream ranking, while richer descriptive annotations are best for upstream retrieval. ALL COMBINED loses across every metric at every K , confirming that uncured metadata harms both stages.

7 Conclusion

In this work, we proposed a multimodal framework for sequential music recommendation that enriches traditional ID-based recommendation pipelines with semantic metadata, audio representations, lyric embeddings, and behavioral engagement signals. We constructed a large-scale multimodal benchmark on top of LastFM-1K and evaluated multiple recommendation backbones, embedding models, and fusion strategies under both zero-shot and fine-tuned settings.

Our experiments show that multimodal enrichment improves recommendation quality over ID-only baselines, with lyric embeddings providing the most consistent gains across models and evaluation metrics. In contrast, audio embeddings exhibit less stable improvements across backbones, highlighting the challenges of aligning acoustic representations with LLM-based recommendation architectures. We further observe that naive multimodal fusion does not consistently yield additive gains, suggesting that effective cross-modal alignment remains an open problem for multimodal recommendation systems.

Overall, our findings demonstrate the importance of semantically grounded item representations for music recommendation beyond collaborative interaction signals alone. Future work includes exploring adaptive fusion mechanisms, stronger cross-modal alignment techniques, end-to-end multimodal training, multilingual recommendation settings, and richer user engagement modeling.

Table 10: Zero-shot results for Llama-2-13B.

LLM Backbone	Item Backbone	Configuration	@5		@10		@20	
			Recall	NDCG	Recall	NDCG	Recall	NDCG
LLaMA-2-13B	SASRec	ItemID only	0.030	0.016	0.061	0.026	0.127	0.042
		ItemID Only + CR (Embed)	0.022	0.012	0.053	0.022	0.094	0.032
		ItemID Only + CR (Prompt)	0.025	0.015	0.049	0.022	0.099	0.035
		+ Audio (Concat)	0.025	0.016	0.056	0.026	0.107	0.039
		+ Audio (Weighted Avg.)	0.019	0.009	0.049	0.018	0.096	0.030
		+ Audio (Cross-Attn.)	0.025	0.015	0.058	0.025	0.096	0.035
		+ Audio (FiLM)	0.025	0.016	0.044	0.021	0.086	0.032
		+ Lyric (Concat)	0.022	0.015	0.041	0.021	0.100	0.036
		+ Lyric (Weighted Avg.)	0.017	0.010	0.036	0.016	0.080	0.027
		+ Lyric (Cross-Attn.)	0.020	0.012	0.047	0.020	0.102	0.034
		+ Lyric (FiLM)	0.016	0.008	0.034	0.015	0.075	0.025
		+ Audio + Lyric + CR (Concat - Prompt)	0.030	0.016	0.058	0.025	0.096	0.035
		+ Audio + Lyric(Concat) + CR(Embed)	0.028	0.019	0.055	0.027	0.107	0.040
		+ Audio + Lyric (Concat)	0.024	0.015	0.050	0.023	0.102	0.037
		+ Audio + Lyric (Weighted Avg.)	0.028	0.015	0.050	0.022	0.103	0.036
		+ Audio + Lyric (Cross-Attn.)	0.019	0.010	0.045	0.019	0.075	0.026
		+ Audio + Lyric (FiLM)	0.024	0.013	0.049	0.021	0.091	0.032
	BERT4Rec	ItemID only	0.024	0.014	0.056	0.025	0.108	0.038
		ItemID Only + CR (Embed)	0.027	0.016	0.053	0.025	0.110	0.039
		ItemID Only + CR (Prompt)	0.031	0.017	0.064	0.027	0.129	0.044
		+ Audio (Concat)	0.033	0.017	0.053	0.024	0.108	0.037
		+ Audio (Weighted Avg.)	0.009	0.005	0.034	0.013	0.075	0.024
		+ Audio (Cross-Attn.)	0.033	0.020	0.060	0.028	0.102	0.039
		+ Audio (FiLM)	0.034	0.020	0.061	0.028	0.122	0.044
		+ Lyric (Concat)	0.036	0.022	0.060	0.029	0.118	0.113
		+ Lyric (Weighted Avg.)	0.028	0.014	0.050	0.021	0.103	0.035
		+ Lyric (Cross-Attn.)	0.017	0.011	0.047	0.020	0.082	0.029
		+ Lyric (FiLM)	0.016	0.007	0.036	0.014	0.096	0.029
		+ Audio + Lyric + CR (Concat - Prompt)	0.020	0.012	0.056	0.024	0.097	0.034
		+ Audio + Lyric (Concat) + CR (Embed)	0.031	0.018	0.053	0.025	0.097	0.036
		+ Audio + Lyric (Concat)	0.038	0.021	0.063	0.029	0.122	0.044
		+ Audio + Lyric (Weighted Avg.)	0.024	0.016	0.042	0.022	0.088	0.033
		+ Audio + Lyric (Cross-Attn.)	0.027	0.016	0.063	0.027	0.108	0.039
		+ Audio + Lyric (FiLM)	0.020	0.011	0.056	0.022	0.108	0.035
	GRU4Rec	ItemID only	0.030	0.019	0.061	0.029	0.125	0.045
		ItemID Only + CR (Embed)	0.034	0.023	0.060	0.031	0.125	0.048
		ItemID Only + CR (Prompt)	0.038	0.025	0.058	0.032	0.113	0.045
		+ Audio (Concat)	0.027	0.013	0.045	0.019	0.102	0.033
		+ Audio (Weighted Avg.)	0.019	0.010	0.047	0.020	0.108	0.035
		+ Audio (Cross-Attn.)	0.020	0.012	0.042	0.019	0.092	0.031
		+ Audio (FiLM)	0.031	0.022	0.064	0.032	0.121	0.047
		+ Lyric (Concat)	0.028	0.016	0.052	0.023	0.089	0.032
		+ Lyric (Weighted Avg.)	0.024	0.014	0.042	0.020	0.089	0.032
		+ Lyric (Cross-Attn.)	0.024	0.014	0.060	0.025	0.094	0.033
		+ Lyric (FiLM)	0.036	0.021	0.061	0.029	0.110	0.041
		+ Audio + Lyric + CR (Concat - Prompt)	0.024	0.015	0.044	0.021	0.097	0.035
		+ Audio + Lyric (Concat) + CR (Embed)	0.020	0.012	0.049	0.021	0.113	0.037
		+ Audio + Lyric (Concat)	0.016	0.010	0.049	0.020	0.108	0.035
		+ Audio + Lyric (Weighted Avg.)	0.022	0.014	0.053	0.024	0.114	0.039
		+ Audio + Lyric (Cross-Attn.)	0.014	0.007	0.049	0.018	0.100	0.031
		+ Audio + Lyric (FiLM)	0.020	0.015	0.041	0.021	0.080	0.031

Table 11: Fine-tuned results for Llama-2-13B (SASRec backbone).

LLM Backbone	Item Backbone	Configuration	@5		@10		@20	
			Recall	NDCG	Recall	NDCG	Recall	NDCG
LLaMA-2-13B-Finetuned	SASRec	ItemID only	0.229	0.125	0.323	0.155	0.398	0.174
		+ Audio (Concat)	0.223	0.122	0.323	0.154	0.401	0.174
		+ Audio (Weighted Avg.)	0.232	0.124	0.317	0.151	0.397	0.172
		+ Audio (Cross-Attn.)	0.240	0.130	0.326	0.158	0.395	0.175
		+ Audio (FiLM)	0.232	0.128	0.334	0.161	0.398	0.177
		+ Lyric (Concat)	0.229	0.127	0.321	0.157	0.395	0.176
		+ Lyric (Weighted Avg.)	0.208	0.114	0.317	0.149	0.390	0.168
		+ Lyric (Cross-Attn.)	0.227	0.123	0.321	0.154	0.400	0.174
		+ Lyric (FiLM)	0.215	0.118	0.298	0.145	0.387	0.168
		+ Audio + Lyric (Concat)	0.234	0.127	0.317	0.154	0.404	0.177
		+ Audio + Lyric (Weighted Avg.)	0.230	0.125	0.313	0.152	0.395	0.173
		+ Audio + Lyric (FiLM)	0.241	0.131	0.326	0.158	0.395	0.176

Table 12: Zero-shot results for LLaMA-3-70B.

LLM Backbone	Item Backbone	Configuration	@5		@10		@20	
			Recall	NDCG	Recall	NDCG	Recall	NDCG
LLaMA-3-70B	SASRec	ItemID only	0.022	0.015	0.049	0.023	0.100	0.036
		ItemID Only + CR (Embed)	0.020	0.009	0.038	0.015	0.075	0.024
		ItemID Only + CR (Prompt)	0.025	0.019	0.052	0.028	0.100	0.040
		+ Audio (Concat)	0.027	0.013	0.053	0.021	0.086	0.029
		+ Audio (Weighted Avg.)	0.022	0.012	0.050	0.022	0.113	0.037
		+ Audio (Cross-Attn.)	0.028	0.018	0.063	0.029	0.116	0.042
		+ Audio (FiLM)	0.030	0.022	0.061	0.032	0.113	0.045
		+ Lyric (Concat)	0.020	0.013	0.049	0.022	0.091	0.032
		+ Lyric (Weighted Avg.)	0.020	0.011	0.033	0.015	0.086	0.029
		+ Lyric (Cross-Attn.)	0.025	0.018	0.042	0.023	0.089	0.035
		+ Lyric (FiLM)	0.022	0.012	0.049	0.021	0.096	0.032
		+ Audio + Lyric (Concat)	0.038	0.023	0.061	0.031	0.114	0.044
		+ Audio + Lyric (Weighted Avg.)	0.028	0.015	0.060	0.025	0.111	0.038
		+ Audio + Lyric (Cross-Attn.)	0.022	0.013	0.042	0.020	0.094	0.033
		+ Audio + Lyric (FiLM)	0.011	0.006	0.045	0.017	0.089	0.027
		<i>Best overall (SASRec)</i>	0.038	0.023	0.063	0.032	0.116	0.045
	BERT4Rec	ItemID only	0.020	0.014	0.042	0.021	0.089	0.033
		ItemID Only + CR (Embed)	0.017	0.010	0.045	0.019	0.086	0.029
		ItemID Only + CR (Prompt)	0.038	0.024	0.056	0.030	0.092	0.039
		+ Audio (Concat)	0.027	0.014	0.042	0.019	0.096	0.033
		+ Audio (Weighted Avg.)	0.020	0.012	0.045	0.020	0.096	0.032
		+ Audio (Cross-Attn.)	0.028	0.016	0.053	0.024	0.091	0.033
		+ Audio (FiLM)	0.022	0.013	0.049	0.021	0.099	0.034
		+ Lyric (Concat)	0.014	0.007	0.036	0.014	0.082	0.026
		+ Lyric (Weighted Avg.)	0.024	0.016	0.041	0.022	0.094	0.035
		+ Lyric (Cross-Attn.)	0.024	0.017	0.047	0.025	0.108	0.040
		+ Lyric (FiLM)	0.024	0.015	0.053	0.024	0.108	0.038
		+ Audio + Lyric + CR (Concat-Prompt)	0.019	0.013	0.044	0.021	0.092	0.034
		+ Audio + Lyric (Concat) + CR (Embed)	0.024	0.015	0.041	0.020	0.099	0.035
		+ Audio + Lyric (Concat)	0.033	0.022	0.075	0.036	0.124	0.048
		+ Audio + Lyric (Weighted Avg.)	0.025	0.016	0.053	0.025	0.105	0.037
		+ Audio + Lyric (Cross-Attn.)	0.034	0.022	0.056	0.029	0.102	0.040
		+ Audio + Lyric (FiLM)	0.024	0.013	0.049	0.021	0.078	0.029
	GRU4Rec	ItemID only	0.025	0.015	0.044	0.021	0.088	0.032
		ItemID Only + CR (Embed)	0.027	0.015	0.064	0.027	0.118	0.040
		ItemID Only + CR (Prompt)	0.024	0.013	0.055	0.023	0.127	0.041
		+ Audio (Concat)	0.025	0.014	0.050	0.022	0.100	0.035
		+ Audio (Weighted Avg.)	0.024	0.015	0.042	0.021	0.080	0.030
		+ Audio (Cross-Attn.)	0.025	0.015	0.042	0.021	0.097	0.034
		+ Audio (FiLM)	0.028	0.014	0.045	0.019	0.096	0.032
		+ Lyric (Concat)	0.034	0.019	0.056	0.026	0.100	0.037
		+ Lyric (Weighted Avg.)	0.027	0.014	0.049	0.022	0.119	0.039
		+ Lyric (Cross-Attn.)	0.017	0.010	0.033	0.015	0.086	0.028
		+ Lyric (FiLM)	0.008	0.005	0.036	0.014	0.097	0.029
		+ Audio + Lyric + CR (Concat-Prompt)	0.025	0.018	0.052	0.027	0.096	0.038
		+ Audio + Lyric (Concat) + CR (Embed)	0.019	0.013	0.038	0.019	0.099	0.034
		+ Audio + Lyric (Concat)	0.025	0.016	0.045	0.022	0.092	0.034
		+ Audio + Lyric (Weighted Avg.)	0.019	0.012	0.045	0.020	0.100	0.034
		+ Audio + Lyric (Cross-Attn.)	0.025	0.017	0.053	0.026	0.116	0.041
		+ Audio + Lyric (FiLM)	0.025	0.014	0.052	0.022	0.108	0.036

Table 13: Per-LLM aggregates over all 30 (item-ID backbone $\times$ config) cells.

LLM	NDCG@10 mean	NDCG@10 max	Recall@10 mean	Recall@20 mean
Qwen2.5-7B	0.018 $\pm$ 0.003	0.025	0.046	0.096
Llama-2-7B	0.018 $\pm$ 0.003	0.028	0.046	0.097
Llama-3-70B	0.018 $\pm$ 0.003	0.024	0.045	0.097

Table 14: Best metadata configuration per (LLM, item-ID backbone) cell, ranked by NDCG@10.

LLM	Backbone	Best config	NDCG@10	Recall@10	Recall@20
Qwen2.5-7B	SASRec	All with signal + CR	0.025	0.058	0.107
	BERT4Rec	Best with signal + CR	0.024	0.058	0.099
	GRU4Rec	All Spotify + CR	0.019	0.047	0.102
Llama-2-7B	SASRec	All Spotify + CR	0.023	0.056	0.111
	BERT4Rec	All with signal + CR	0.028	0.066	0.107
	GRU4Rec	All with signal	0.024	0.058	0.100
Llama-3-70B	SASRec	All MGPhot	0.024	0.063	0.097
	BERT4Rec	All Combined + CR	0.022	0.055	0.102
	GRU4Rec	All with signal + CR	0.024	0.058	0.118

Table 15: Zero-shot NDCG@10 across all 90 (LLM $\times$ item-ID backbone $\times$ metadata config) cells on LastFM. Best per row bolded.

Config	Qwen2.5-7B			Llama-2-7B			Llama-3-70B		
	SASRec	BERT4Rec	GRU4Rec	SASRec	BERT4Rec	GRU4Rec	SASRec	BERT4Rec	GRU4Rec
All Spotify	0.022	0.016	0.015	0.019	0.020	0.019	0.018	0.013	0.021
All Spotify + CR	0.017	0.015	0.019	0.023	0.019	0.019	0.019	0.014	0.019
All MGPhot	0.023	0.021	0.016	0.013	0.018	0.018	0.024	0.014	0.015
All MGPhot + CR	0.013	0.020	0.016	0.016	0.019	0.021	0.018	0.015	0.013
All Combined	0.010	0.014	0.018	0.019	0.016	0.014	0.016	0.017	0.014
All Combined + CR	0.016	0.018	0.016	0.014	0.012	0.016	0.017	0.022	0.019
All with signal	0.020	0.019	0.019	0.020	0.021	0.024	0.022	0.017	0.019
All with signal + CR	0.025	0.019	0.017	0.019	0.028	0.014	0.019	0.018	0.024
Best with signal	0.017	0.019	0.019	0.022	0.015	0.015	0.022	0.019	0.016
Best with signal + CR	0.022	0.024	0.018	0.016	0.019	0.018	0.017	0.018	0.015

Table 16: Zero-shot Recall@20 across all 90 (LLM $\times$ item-ID backbone $\times$ metadata config) cells on LastFM. Best per row bolded.

Config	Qwen2.5-7B			Llama-2-7B			Llama-3-70B		
	SASRec	BERT4Rec	GRU4Rec	SASRec	BERT4Rec	GRU4Rec	SASRec	BERT4Rec	GRU4Rec
All Spotify	0.103	0.086	0.094	0.089	0.092	0.077	0.085	0.089	0.099
All Spotify + CR	0.088	0.092	0.102	0.111	0.107	0.096	0.097	0.085	0.099
All MGPhot	0.121	0.122	0.089	0.085	0.099	0.099	0.097	0.088	0.092
All MGPhot + CR	0.078	0.114	0.091	0.096	0.100	0.119	0.099	0.097	0.091
All Combined	0.086	0.085	0.096	0.107	0.100	0.086	0.089	0.097	0.100
All Combined + CR	0.080	0.091	0.103	0.097	0.074	0.099	0.096	0.102	0.114
All with signal	0.108	0.113	0.086	0.105	0.107	0.100	0.118	0.078	0.092
All with signal + CR	0.107	0.094	0.103	0.083	0.107	0.083	0.118	0.092	0.118
Best with signal	0.092	0.088	0.099	0.118	0.094	0.085	0.099	0.103	0.097
Best with signal + CR	0.089	0.099	0.094	0.094	0.099	0.092	0.086	0.096	0.088

Table 17: Zero-shot MRR@10 across all 90 (LLM $\times$ item-ID backbone $\times$ metadata config) cells on LastFM. Best per row bolded.

Config	Qwen2.5-7B			Llama-2-7B			Llama-3-70B		
	SASRec	BERT4Rec	GRU4Rec	SASRec	BERT4Rec	GRU4Rec	SASRec	BERT4Rec	GRU4Rec
All Spotify	0.013	0.009	0.008	0.011	0.011	0.012	0.010	0.007	0.011
All Spotify + CR	0.009	0.008	0.011	0.013	0.010	0.011	0.010	0.007	0.009
All MGPhot	0.013	0.012	0.010	0.008	0.010	0.010	0.012	0.008	0.008
All MGPhot + CR	0.007	0.011	0.008	0.008	0.010	0.012	0.010	0.008	0.007
All Combined	0.005	0.008	0.010	0.011	0.008	0.008	0.008	0.009	0.008
All Combined + CR	0.009	0.010	0.009	0.008	0.006	0.009	0.008	0.013	0.012
All with signal	0.010	0.010	0.011	0.010	0.011	0.013	0.012	0.009	0.011
All with signal + CR	0.015	0.010	0.008	0.010	0.016	0.007	0.011	0.010	0.014
Best with signal	0.009	0.011	0.010	0.012	0.009	0.008	0.012	0.010	0.009
Best with signal + CR	0.012	0.013	0.010	0.008	0.010	0.011	0.009	0.010	0.008

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A MGPHot Annotation Validation

To assess the fidelity of our LLM-generated MGPHot annotations, we compare them against the published MGPHot ground-truth ratings [23] on the intersection of our top-50k LastFM song list and the MGPHot release (inner join on the artist/title pair). For each of the 58 MGPHot attributes, we compute Spearman’s rank correlation ρ (rank-order agreement between predicted and ground-truth ratings) and mean absolute error (MAE) on the $[0, 1]$ -normalized scale.

Both metrics are evaluated for the `no_meta_few_shot` prompting variant described in Section 3.2, which is the prompt that produced the annotations used throughout the paper.

Figures 5–11 report per-attribute results, grouped by the seven MGPHot categories of Table 2. Across all 58 attributes, Spearman correlations are positive and MAE values are uniformly below 0.25 on the $[0, 1]$ scale, indicating that the LLM-generated annotations preserve the rank-order of the ground-truth ratings while staying within a tight absolute-error band. Agreement is strongest for lyric mood dimensions (e.g., $\rho \approx 0.72$ for Love/Romance Lyrics, 0.67 for Sad Lyrics) and weakest for perceptually subtle vocal attributes (e.g., $\rho \approx 0.27$ for Vocal Nasality), consistent with the intuition that lexically grounded properties are easier for an LLM to infer from title-and-artist context than fine-grained acoustic ones.

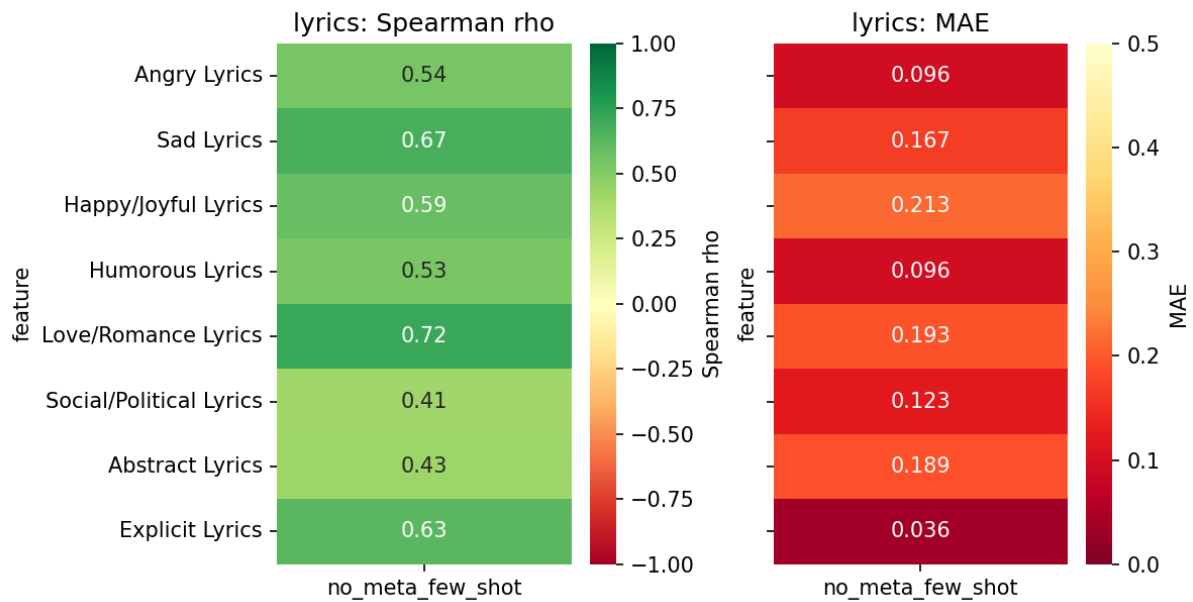


Figure 5: MGPHot validation – Lyrics category. Left: Spearman's ρ between LLM-generated and ground-truth ratings; right: MAE on $[0, 1]$.

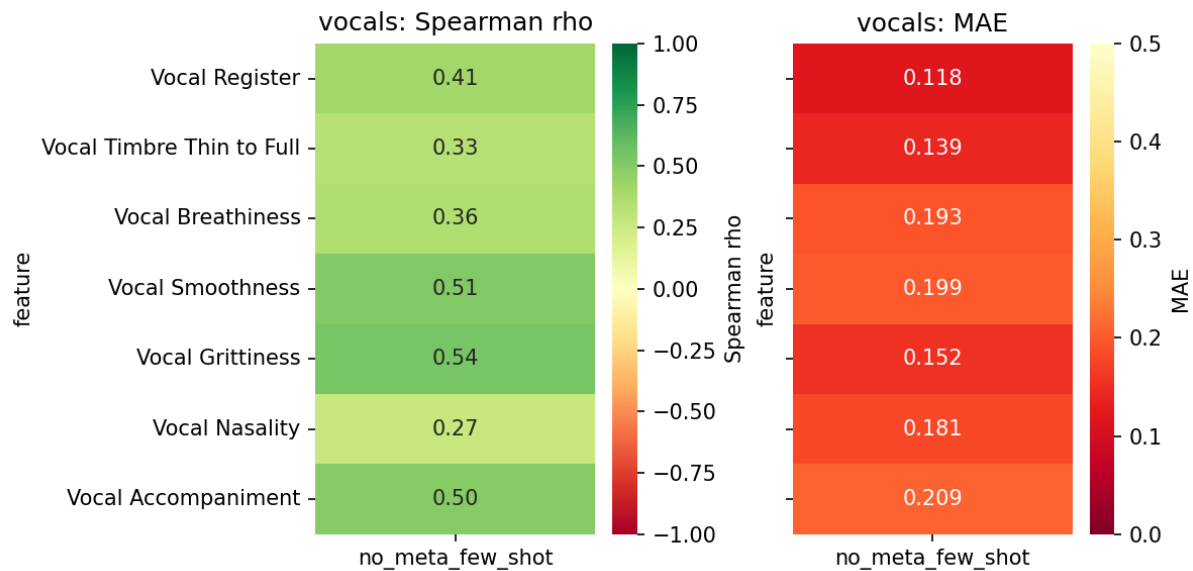


Figure 6: MGPHot validation – Vocals category. Left: Spearman's ρ ; right: MAE.

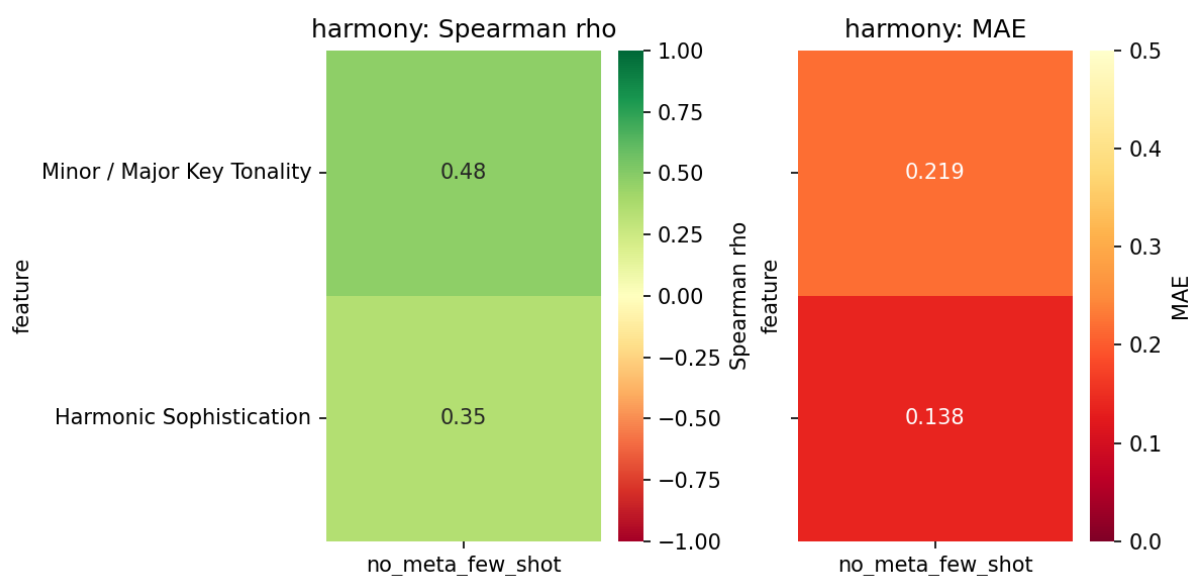


Figure 7: MGPHot validation – Harmony category. Left: Spearman's ρ ; right: MAE.

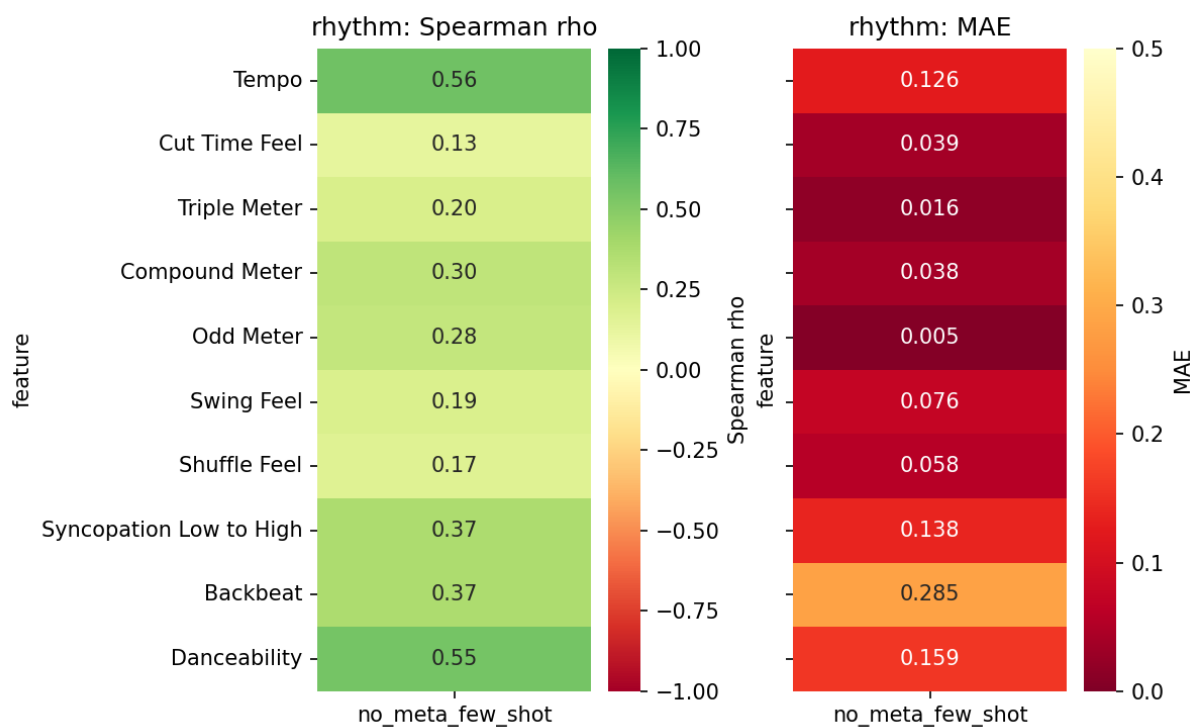


Figure 8: MGPHot validation – Rhythm category. Left: Spearman's ρ ; right: MAE.

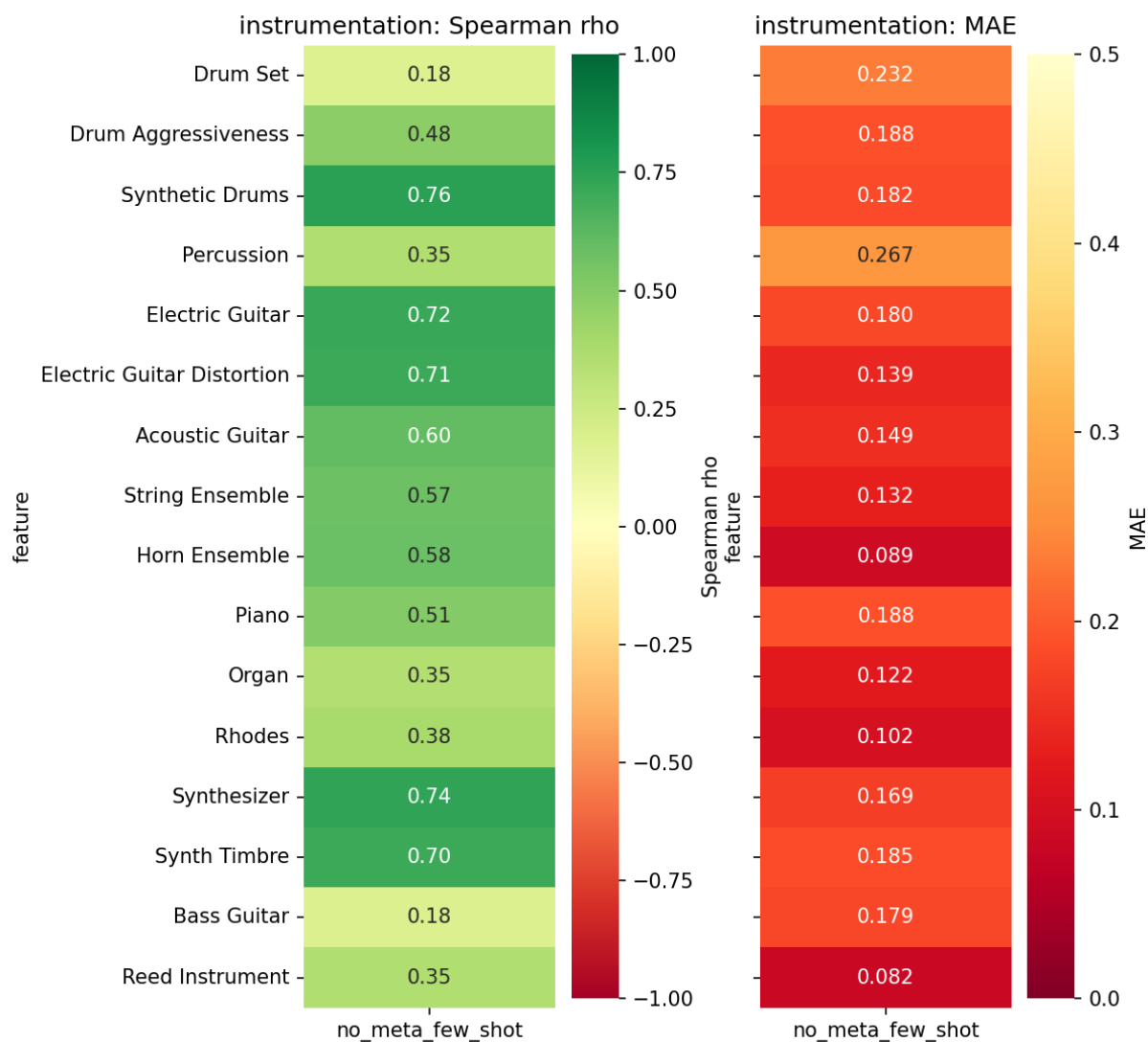


Figure 9: MGPHot validation – Instrumentation category. Left: Spearman's ρ ; right: MAE.

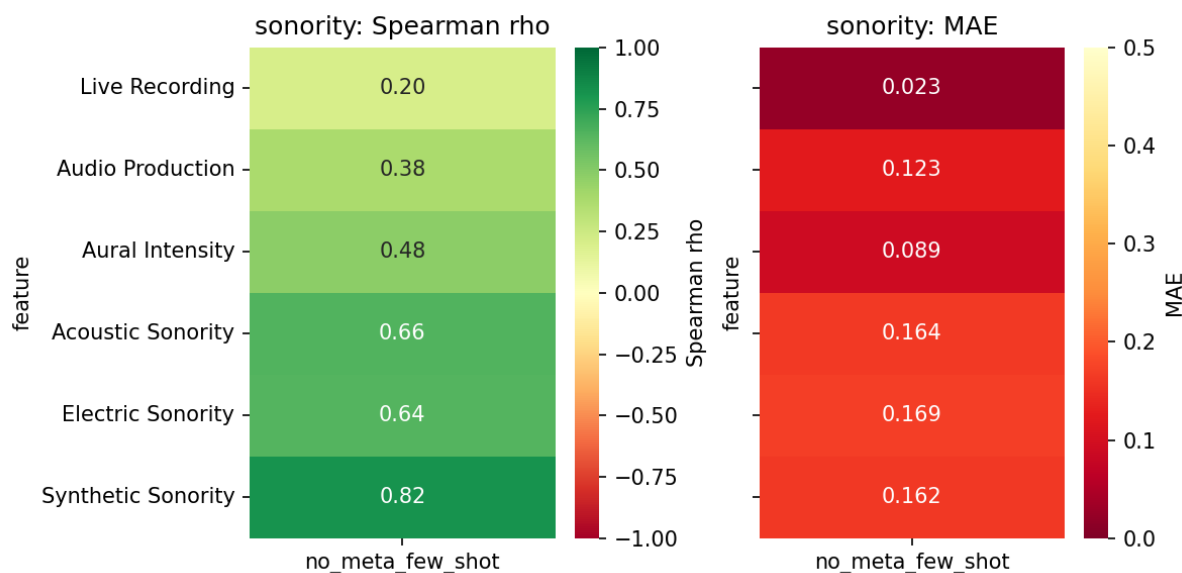


Figure 10: MGPHot validation – Sonority category. Left: Spearman's ρ ; right: MAE.

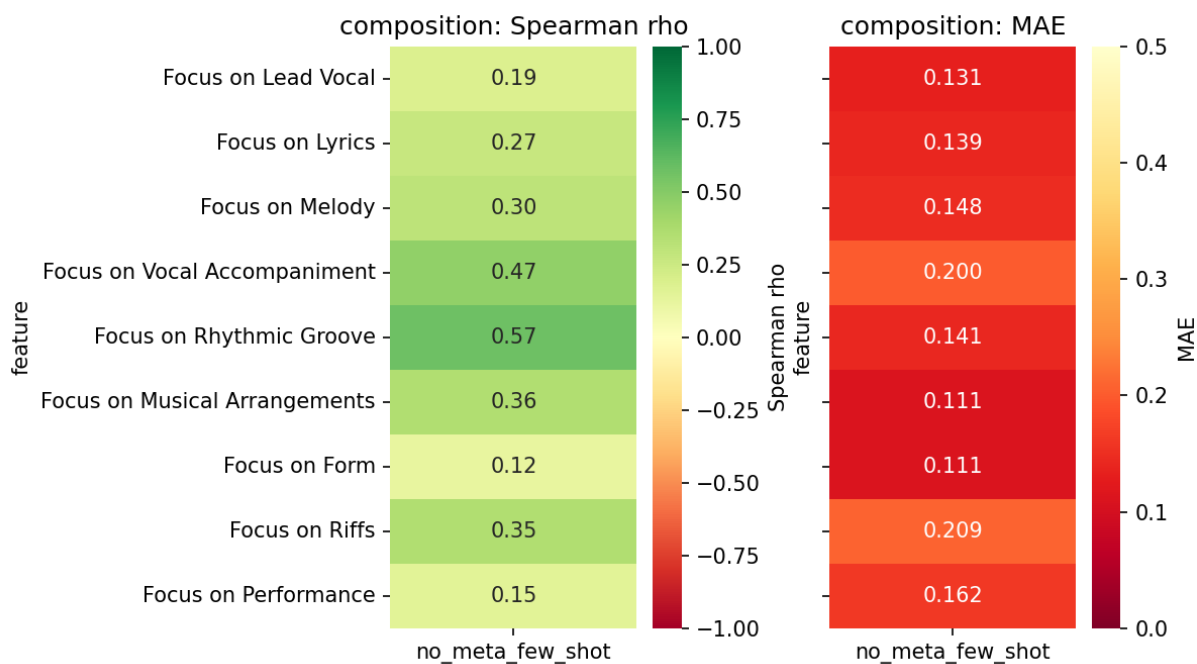


Figure 11: MGPHot validation – Composition category. Left: Spearman's ρ ; right: MAE.